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To train or not to train (on match day)? Influence of a priming session on match performance in competitive elite-level soccer

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ABSTRACT

This study examined the effect of a morning priming session consisted of stretching, mobility, core and lower-body resistance, and reactive agility exercises on match performance in elite-level soccer players. Information on physical and technical performance was collected during competitive matches ($n = 32$) and subsequently classified as those preceded or not preceded by a priming session. Results showed significantly increased (i) distance covered overall (Cohen's d (d) = 0.34, $p = 0.011$), in moderate-intensity (d = 0.52, $p = 0.001$) and high-intensity running (d = 0.30, $p = 0.024$), and (ii) frequency of duels (d = 0.26, $p = 0.050$) in matches preceded by a priming session compared to those not preceded by a priming session. These findings show that players' physical performance was notably enhanced due to the pre-match routines that the team implemented on match day, while technical performance was in general similar irrespective of whether a priming session was implemented or not before the match. Therefore, morning priming session implemented on match day can be recommended as an effective pre-match strategy to increase players' physical performance in soccer without negative consequences on technical performance.

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Match-day training; physical performance; technical performance; football

Introduction

Match analysis is commonly used in many sports as it provides a means for coaches to collect objective information which can be used to provide feedback on match performance (Kim et al., 2023). In soccer, match performance is determined by a complex interaction between physical, technical, and tactical performance and psycho-social factors (Schuth et al., 2016). Physical performance is commonly evaluated by quantification of different kinematic variables including total distance covered, distances covered in different speed zones, and acceleration frequency (Byrkjedal et al., 2023; Riboli & Castagna, 2023). Shots, passes, crosses, and dribbles are among the most frequently used variables to evaluate technical performance (Yi et al., 2019), while inter-player coordination and inter-team coordination are often analysed to evaluate tactical performance (Folgado et al., 2015). Research investigating performance in soccer match-play highlighted players' technical performance as decisive for competitive success (Hoppe et al., 2015; Moalla et al., 2018). However, achieving high level of soccer match-play required in contemporary soccer (Barnes et al., 2014) might not be possible without an appropriate level of physical performance (Mackenzie & Cushion, 2013). Therefore, enhancing players' technical and physical performance constitutes an essential objective for soccer practitioners.

Together with appropriate recovery and nutrition, an important strategy to enhance technical and physical performance on match day is the periodisation of training load (Jeong et al., 2011; Los Arcos et al., 2017). Most of the efforts aimed to optimise match preparation are performed during the weekly training sessions; however, the final preparation takes part on match day. Thus, depending on the strategy of the coaching staff, players on the morning of match day (i.e., as matches are typically played in the afternoon or evening) can perform "priming" session (i.e., in soccer, so-called "activation"). Research investigating the prevalence and application of priming exercises in high-performance sports shows that priming sessions typically consist of upper-body (i.e., bench press, medicine ball throw, chin up, etc.) and lower-body exercises (i.e., squat, Olympic lift variations, jumps, etc.), usually involving both loaded and unloaded activities (Harrison et al., 2020).

Priming sessions in general aim to enhance afternoon/evening performance through several mechanisms such as reducing joint and muscle stiffness, high-frequency motor neuron activation, and acute elevation of testosterone (Harrison et al., 2019; Nutt et al., 2022). However, while some empirical evidence suggests that priming sessions may be an effective strategy to enhance athletes' performance (Marrier et al., 2019; Mason et al., 2017; McGowan et al., 2017; Nutt et al., 2022), contrasting findings have reported no positive effect of priming sessions on performance (Fry et al., 1995; Harrison et al., 2020). These inconsistencies observed across studies may be due in part to the different approaches used in the prescription of priming sessions, as a wide range of methods

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have been applied in practice (Harrison et al., 2020). Irrespective of causality, it is important to highlight that previous studies have investigated only instantaneous measures of performance such as jumping and/or strength indicators.

Despite the demonstrated importance of physical and technical performance in soccer (Hoppe et al., 2015; Mackenzie & Cushion, 2013; Moalla et al., 2018; Schuth et al., 2016), to the best of our knowledge, no study has examined the effects of priming sessions on these performances in official soccer matches so far (Harrison et al., 2020). Consequently, it is currently unclear if priming sessions may have or not a positive effect on soccer players performance during the competition, and additional research is warranted. The findings from such research will provide novel knowledge that can be used by soccer practitioners to enhance players' match performance. Therefore, this study aimed to examine the effect of priming sessions on the physical and technical performance of elite-level players in competition.

Materials and methods

Experimental approach

In this observational study, the match performance of soccer players was analysed according to two specific routines that the team implemented on match day: (i) a 15–20 minute priming session, (ii) a 15–20 minute period of walking, or resting. This design allowed for the potential differences in performances in matches preceded and not preceded by priming sessions to be investigated. Comprehensive empirical evidence has shown the influence of playing position, match location and outcome, quality of the opponent, and type of the pitch on match performance (Zhou et al., 2021). As such, these variables were considered contextual factors that needed to be statistically controlled for and were thus included as covariates (i.e., independent variables).

Sample

Data were collected from 31 outfield professional players from the senior team of a professional soccer club. Nine were central midfielders (CM), seven were central defenders (CD), four were forwards (FW), five were fullbacks (FB), and six were wide midfielders (WM). Players participated in 32 official matches in the Latvian first-division. Of these, 12 matches were preceded by a priming session and 20 matches were not preceded by a priming session. No red cards were incurred during the matches. Only matches played in a customary 4-3-3 formation were included in the analysis to avoid the possible effect of changes in team formation on match performance (Modric et al., 2022). Individual observations were included if the player participated in at least 75-minutes of play occurred, as this minimum timeframe can be extrapolated to the 90-minute playtime of a typical game following previously established guidelines (Stevens et al., 2017). In total, 266 individual observations were obtained. Of these, 92 observations were preceded by a priming session and 174 observations were not preceded by a priming session.

The data analysed in this study were derived from player activities that were routinely measured over the course of the competitive season and therefore written informed consent was not required (Rago et al., 2021). However, verbal consent was obtained from the club and the players, and anonymity was ensured. All data were de-identified in accordance with the declaration of Helsinki to ensure confidentiality. The investigation was approved by the local university ethics board (approval number: 2181-205-02-05-19-0020)

Data processing

The players' physical performance was measured using a 10-Hz global positioning system (GPS) (Vector S7, Catapult Sports Ltd., Melbourne, Australia). Previous investigations have demonstrated this system to be valid and reliable for use in applied sport-research settings (i.e., less than 1% measurement error and 80% of common variance with running speed measured by timing gates) (Castellano, 2011; Johnston et al., 2014). To avoid inter-unit error, the players wore the same device for each match. In accordance with the manufacturer's instructions, all devices were consistently activated ~30 minutes before the pre-match warm-up to allow the acquisition of satellite signals. Raw data files were exported after the conclusion of each match using system-specific software (OpenField, Catapult Sports Ltd., Melbourne, Australia). Files were trimmed on an individual player basis to ensure that only data pertaining to each specific match were retained for analysis.

Players' technical performance was investigated using the InStat Scout semi-automatic optical system (Instat Limited, Limerick, Republic of Ireland). The reliability of the system has been recently tested by analysing passes, duels, air duels, tackles, dribbles, crosses, shots, and key passes. All these variables had a bias < 0.2 and limits of agreement were all within 3 and – 3 revealing that this system has high reliability according to the analysed variables (Silva & Marcelino, 2022).

Priming protocol

Considering conflict findings in current research which typically investigated high-intensity work in priming session and its influence on performance (Fry et al., 1995; Harrison et al., 2020; Marrier et al., 2019; Mason et al., 2017; McGowan et al., 2017; Nutt et al., 2022), new protocol, based on practical expertise of the authors of this study who have demonstrated history of working in elite-level soccer as strength and conditioning and soccer coaches, was constructed (Figure 1). The protocol lasted 15–20 minutes and consisted of four parts: (i) ground-floor warm-up, (ii) core resistance exercises, (iii) lower-body resistance exercises, and (iv) reactive agility drills. The first part lasted ~5 minutes and included dynamic exercises aimed to improve muscle flexibility (i.e., 90/90 hamstring stretch, side-lying dynamic quad stretch, back roll to toe touch, etc.) and joint mobility (i.e., rocking frog, half kneeling dorsiflexion mobilisation, lying trunk rotations, etc.). Whole-body exercises were applied, but the focus was on the lower body and core muscles. All exercises were performed in 3–5 repetitions x 1 set, with no rest between exercises. The second part lasted ~5 minutes and included 6–10 body-weight core exercises (i.e., different

variations of front, side, and back planks). One set per exercise was performed, with both a work and rest time of 20 seconds. The third part lasted ~5 minutes and included 10–15 unloaded lower body exercises (i.e., different variations of squats, lunges, deadlifts, and steps). After each exercise was performed, players performed a moderate pace a run over 10 metres. All exercises were performed in 3–5 repetitions x 1 set, with resting time between exercises of ~20 seconds. In the fourth part, which lasted ~3 minutes, reactive agility drills were performed. Players had to react maximally on an unpredictable visual or sound signal given by the coach and subsequently perform a 5 metres maximal acceleration. Two drills were performed, separated by a ~60 seconds rest interval. This protocol was performed on match day mornings and managed by the team's strength and conditioning coach (one of the co-authors of this study). Depending on the time of the match, priming sessions were held 6–9 hours before the match.

Variables

All the physical and technical performance variables with their associated definitions are presented in Table 1. The contextual variables included a five-level categorical variable "playing position", a three-level categorical variable "match outcome", and two-level categorical variables "match location", "quality of the opponent", and "type of the pitch". Playing position classification was done based on players' activity on the pitch and the primary area where this activity was performed and included CD, FB, CM, WM, or FW. The match outcome was assessed as "won", "loss" or "draw". Match location was recorded as playing at "home" or "away". Opponent quality was recorded as "strong" or "weak" for opponents respectively ranked from 1st to 5th or 6th to 10th place. The type of pitch was classified depending on the pitch where matches were played as "artificial turf" or "natural grass". To examine the effect of a priming session on players' physical and technical performance, a two-level categorical variable "priming session" was created, and coded accordingly to whether the match was preceded by a priming session or not.

Statistical analysis

A linear mixed model was used to examine the effect of priming sessions on players' physical and technical performance. The physical and technical variables in matches that preceded or were not preceded by priming sessions were included as dependent variables in the model, and playing position, match outcome, match location, quality of the opponent, and type of pitch were the independent variables included as fixed effects. Players' identities were modelled as random effect. The assumptions of homogeneity and normal distributions of residuals were verified. Cohen's d (d) was used to identify effect sizes (ES) and interpreted as follows: trivial (<0.2), small (≥0.2–0.5), moderate (≥0.5–0.8),

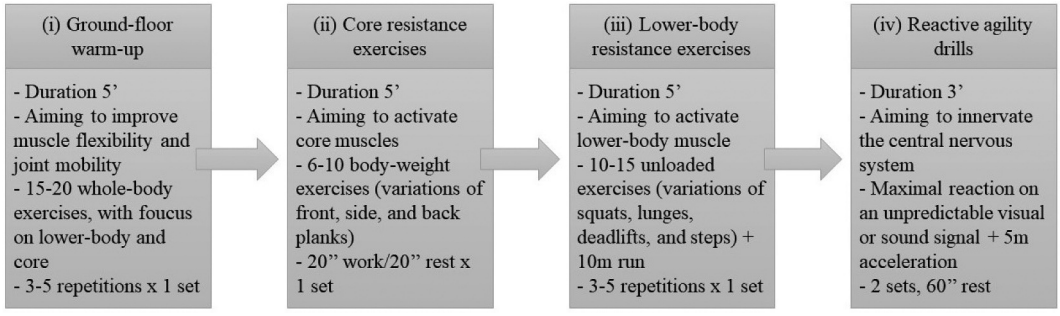

Figure 1. Priming protocol.

Table 1. Physical and technical performance variables and their definitions.

Physical performance	
Total distance (m)	Total distance covered.
Low-intensity running (m)	Distance covered <4 m/s.
Moderate-intensity running (m)	Distance covered 4–5.5 m/s.
High-intensity running (m)	Distance covered 19.8 km/h.
Total accelerations (#)	Accelerations >0.5 m/s^2.
High-intensity accelerations (#)	Accelerations >3 m/s^2.
Total decelerations (#)	Decelerations less than −0.5 m/s^2.
High-intensity decelerations (#)	Decelerations less than −3 m/s^2.
Technical performance	
Shots (#)	An attempt to score a goal, made with any (legal) part of the body, either on or off target.
Passes (#)	An attempt to send ball from one player to another.
Crosses (#)	Attempted passes performed by a player from an offensive zone (last 40 m of pitch between the short side of the penalty area and the lateral side of the field) directly to the penalty area.
Tackles (#)	Active action of a player who tries to tackle the ball from the player possessing it.
Dribbles (#)	Active action of a player possessing the ball who attempted to pass opponent by dribbling the ball.
Duels (#)	Active action of a player which includes the struggle for a neutral ball.

and large (>0.8) (Cohen, 2013). All analyses were performed using SPSS software (IBM, SPSS, version 25.0). Statistical significance was set at $p < 0.05$.

Results

Tables 2 and 3 present descriptive statistics and differences in overall distance and distances covered in different speed zones according to the pre-match routines that the team implemented on match day. In matches preceded by a priming session, players achieved significantly greater total distance (10,727 m versus 10,468 m, respectively; d = 0.34), moderate-intensity running (1632 m versus 1484 m, respectively; d = 0.52), and high-intensity running (689 m versus 625 m, respectively; d = 0.30). As covariates, playing position, match location, and match outcome had an influence on total distance, moderate-intensity running, and high-intensity running. In addition, pitch type had an impact on total distance and high-intensity running.

Tables 2 and 4 present descriptive statistics and differences in acceleration frequencies according to the routines that the team implemented on match day. When matches were preceded by a priming session, players executed a slightly greater number of high-intensity accelerations (30 versus 28, respectively; d = 0.24) and decelerations (39 versus 36, respectively; d = 0.24) compared with matches not preceded by a priming session. As covariates, playing position and pitch type had an influence on high-intensity accelerations and decelerations. In addition, match outcome had an impact on high-intensity decelerations.

Tables 2 and 5 present descriptive statistics and differences in technical performance according to the routines that the team implemented on match day. Players performed more duels in matches preceded by a priming session compared to the matches not preceded by a priming session (18.66 and 16.71, respectively; d = 0.26). As covariates, playing position and opponent quality had an influence on duels.

Discussion

This study aimed to examine the effect of pre-match priming sessions on the performance of professional level players during competitive soccer matches while controlling statistically for the possible influence of contextual factors. The findings show that players' (i) physical performance was dependent on the pre-match routines that the team implemented on match day, and (ii) technical performance was similar irrespective of whether a priming session was implemented or not before the match.

Although physical performance in competition is not proven to be a decisive factor in winning matches (Hoppe et al., 2015; Moalla et al., 2018), it nevertheless plays an essential role in soccer. High-intensity efforts are notably important as they relate to activities that can be associated with influencing match outcome such as movements to win the ball and actions to beat opposition players (DiSalvo et al., 2009; McBurnie et al., 2022). As such, it is not surprising that factors able to enhance physical performance in soccer, such as a half-time re-warm up (Williams, 2018) or improvements in players' aerobic performance (Modric et al., 2021), have been investigated. The current study has identified another important strategy to enhance physical performance during soccer matches. Findings show that a priming session implemented on morning match day may enhance players' physical performance during the ensuing competitive match. Players notably covered significantly greater distances overall (small ES) and in moderate-intensity (moderate ES), and high-intensity running (small ES). Slight improvements, though borderline significance, in high-intensity accelerations (small ES) and decelerations (small ES) were also observed in matches when the priming session was performed in compared to the matches when it was not performed. However, considering the similar values for total accelerations and decelerations irrespective of the match-day routines, the effect of the priming session on these actions can be considered limited.

Table 2. Descriptive statistics for physical and technical performance according to the match day routines.

	Preceded by priming session	Not preceded by priming session
	Mean±SE	Mean±SE
Running distances		
Total distance (m)	10727 ± 120	10468 ± 109
Low-intensity running (m)	8404 ± 94	8358 ± 86
Moderate-intensity running (m)	1632 ± 48	1484 ± 43
High-intensity running (m)	689 ± 32	625 ± 28
Acceleration frequencies		
Total accelerations (#)	330 ± 11	324 ± 10
High-intensity accelerations (#)	30 ± 2	28 ± 2
Total decelerations (#)	313 ± 10	305 ± 10
High-intensity decelerations (#)	39 ± 2	36 ± 2
Technical performances		
Shots (#)	1.29 ± 0.21	1.62 ± 0.17
Passes (#)	50.87 ± 3	56.6 ± 2.58
Crosses (#)	1.95 ± 0.28	1.88 ± 0.22
Tackles (#)	3.77 ± 0.37	3.22 ± 0.31
Duels (#)	18.66 ± 1.15	16.71 ± 1.03
Dribbles (#)	3.25 ± 0.58	3.35 ± 0.53

Table 3. Differences in overall distance and distances covered in different speed zones according to the match day routines.

	Total distance				Low-intensity running			
	Coefficient	SE	p	ES	Coefficient	SE	p	ES
Intercept	10787	167	0.001		8521	128	0.001	
Priming session	259	101	0.011	0.34	46	74	0.535	0.08
Position (CD)	−1018	222	0.001	−1.17	−179	173	0.304	−0.24
Position (CM)	148	193	0.444	0.16	151	147	0.307	0.18
Position (FB)	−493	199	0.014	−0.44	−275	151	0.070	−0.29
Position (FW)	−542	211	0.011	−0.38	−234	158	0.140	−0.21
Match outcome (lost)	−483	108	0.001	−0.59	−416	79	0.001	−0.68
Match outcome (draw)	−282	117	0.016	−0.32	−76	85	0.376	−0.12
Match location (home)	55	80	0.491	0.09	165	59	0.005	0.37
Opponent (strong)	−21	92	0.819	−0.03	−57	67	0.395	−0.12
Pitch type (AT)	599	96	0.001	0.80	108	70	0.125	0.19
	Moderate-intensity running				High-intensity running			
	Coefficient	SE	p	ES	Coefficient	SE	p	ES
Intercept	1505	67	0.001		777	44	0.001	
Priming session	148	42	0.001	0.52	64	28	0.024	0.30
Position (CD)	−472	88	0.001	−1.48	−380	58	0.001	−1.86
Position (CM)	164	78	0.038	0.45	−189	51	0.001	−0.81
Position (FB)	−129	80	0.112	−0.31	−104	53	0.054	−0.40
Position (FW)	−227	86	0.009	−0.40	−105	57	0.068	−0.29
Match outcome (lost)	−61	45	0.182	−0.17	−4	30	0.883	−0.01
Match outcome (draw)	−76	49	0.123	−0.20	−133	32	0.001	−0.54
Match location (home)	−59	33	0.080	−0.23	−51	22	0.022	−0.30
Opponent (strong)	6	38	0.865	0.03	30	25	0.236	0.15
Pitch type (AT)	367	40	0.001	1.17	119	26	0.001	0.57

SE = standard error; ES = Effect size; CD = central defender, CM = central midfielder, FB = fullback, FW = forwards, AT = artificial turf.

Table 4. Differences in acceleration frequencies according to the match day routines.

	Total accelerations				High-intensity accelerations			
	Coefficient	SE	p	ES	Coefficient	SE	p	ES
Intercept	316	13	0.001		30	2	0.001	316
Priming session	7	5	0.189	0.17	2	1	0.071	0.24
Position (CD)	2	16	0.888	0.02	−9	3	0.007	−0.51
Position (CM)	6	13	0.654	0.05	−6	3	0.027	−0.32
Position (FB)	−1	12	0.909	−0.01	−6	3	0.033	−0.29
Position (FW)	−20	13	0.116	−0.20	0	3	0.959	−0.01
Match outcome (lost)	−19	6	0.001	−0.45	0	1	0.707	−0.05
Match outcome (draw)	−9	6	0.135	−0.20	2	1	0.114	0.21
Match location (home)	9	4	0.029	0.29	0	1	0.692	−0.05
Opponent (strong)	7	5	0.146	0.20	1	1	0.271	0.14
Pitch type (AT)	23	5	0.001	0.60	2	1	0.040	0.27
	Total-decelerations				High-intensity decelerations			
	Coefficient	SE	p	ES	Coefficient	SE	p	ES
Intercept	297	12	0.001		39	3	0.001	
Priming session	7	5	0.155	0.19	3	1	0.080	0.24
Position (CD)	−4	15	0.820	−0.03	−15	4	0.001	−0.91
Position (CM)	11	12	0.384	0.12	−4	3	0.222	−0.21
Position (FB)	−4	12	0.734	−0.04	−7	3	0.025	−0.35
Position (FW)	−12	12	0.312	−0.13	2	3	0.469	0.09
Match outcome (lost)	−15	5	0.006	−0.37	0	2	0.991	0.00
Match outcome (draw)	−10	6	0.073	−0.24	1	2	0.656	0.05
Match location (home)	10	4	0.017	0.32	0	1	0.778	−0.04
Opponent (strong)	8	5	0.082	0.22	4	1	0.003	0.39
Pitch type (AT)	20	5	0.001	0.55	0	1	0.753	0.04

SE = standard error; ES = Effect size; CD = central defender, CM = central midfielder, FB = fullback, FW = forwards, AT = artificial turf.

Despite the relatively small magnitude of these differences (i.e., all small to moderate effect sizes), these findings arguably have important practical applications for several reasons. Firstly, players' physical performances were ~ 10% greater when the match was preceded by a priming session. Such enhancement in high-intensity efforts (e.g., high and moderate-intensity running, and high-intensity accelerations and deceleration) may be notable in helping achieve higher levels of physical performance compared to opponents. Therefore, the impact of priming sessions on physical performances should be considered a key aspect for teams seeking improvements in match volume and intensity. Secondly, the impact of the priming session on

Table 5. Differences in technical performance according to the match day routines.

	Shots				Passes				Crosses			
	Coefficient	SE	p	ES	Coefficient	SE	p	ES	Coefficient	SE	p	ES
Intercept	1.88	0.29	0.001		53.75	4.29	0.001		3.34	0.38	0.001	
Priming session	−0.34	0.24	0.152	−0.19	−5.73	3.10	0.066	−0.24	0.08	0.34	0.822	0.03
Position (CD)	−1.21	0.33	0.001	−1.23	23.24	5.28	0.001	1.45	−3.63	0.40	0.001	−4.11
Position (CM)	−0.28	0.32	0.376	−0.26	14.59	4.87	0.004	0.79	−3.14	0.39	0.001	−3.46
Position (FB)	−1.13	0.33	0.001	−1.02	20.31	5.10	0.001	1.01	0.13	0.41	0.747	0.15
Position (FW)	1.45	0.39	0.001	0.82	−18.54	5.69	0.002	−0.63	−3.44	0.50	0.001	−2.13
Match outcome (lost)	0.1	0.25	0.690	0.05	7.5	3.31	0.024	0.29	0.59	0.36	0.105	0.21
Match outcome (draw)	−0.08	0.27	0.775	−0.04	−5.56	3.57	0.121	−0.20	0.54	0.39	0.168	0.18
Match location (home)	0.43	0.18	0.019	0.30	5.9	2.44	0.016	0.31	0.31	0.27	0.247	0.15
Opponent (strong)	−0.49	0.21	0.019	−0.29	−19.17	2.77	0.001	−0.87	−0.57	0.30	0.054	−0.24
Pitch type (AT)	−0.01	0.21	0.957	−0.01	1.82	2.87	0.527	0.08	0.61	0.30	0.046	0.26

	Tackles				Duels				Dribbles			
	Coefficient	SE	p	ES	Coefficient	SE	p	ES	Coefficient	SE	p	ES
Intercept	3.68	0.53	0.001		20.9	1.60	0.001		5.42	0.78	0.001	
Priming session	0.56	0.42	0.182	0.17	1.95	0.99	0.050	0.26	−0.1	0.43	0.818	−0.03
Position (CD)	−0.82	0.61	0.187	−0.44	−6.3	2.12	0.004	−0.77	−3.62	1.06	0.001	−0.85
Position (CM)	0.4	0.58	0.494	0.19	−4.97	1.85	0.008	−0.54	−2.32	0.89	0.010	−0.48
Position (FB)	0.53	0.61	0.384	0.25	−4.64	1.91	0.017	−0.44	−1.97	0.90	0.031	−0.35
Position (FW)	−2.07	0.70	0.004	−0.62	−5.18	2.03	0.012	−0.38	−1.2	0.94	0.202	−0.18
Match outcome (lost)	0.76	0.44	0.090	0.22	1.93	1.05	0.068	0.24	0.85	0.46	0.069	0.24
Match outcome (draw)	0.61	0.48	0.207	0.16	−0.03	1.14	0.976	0.00	−0.83	0.50	0.098	−0.22
Match location (home)	−0.28	0.33	0.389	−0.11	−0.56	0.78	0.472	−0.09	0.16	0.34	0.640	0.06
Opponent (strong)	−0.83	0.37	0.025	−0.28	−1.79	0.89	0.045	−0.26	−1.23	0.39	0.002	−0.41
Pitch type (AT)	0.07	0.38	0.858	0.02	1.14	0.93	0.220	0.16	0.55	0.41	0.180	0.17

SE = standard error; ES = Effect size; CD = central defender, CM = central midfielder, FB = fullback, FW = forwards, AT = artificial turf.

physical performance accounted for different contextual variables which have been demonstrated to affect performance in soccer (Zhou et al., 2021). This approach strengthens the true effect that the present priming sessions had on physical performance in professional soccer players while at the same time rejects possible conclusion that this can be due to the natural match-to-match variability (Carling et al., 2016). Thirdly, typical priming sessions include strength-power exercises such as jumping, squatting, pressing, and pulling derivatives that are often performed at maximal or near-maximal velocities (Harrison et al., 2020). Although effective in enhancing performance, such highly intensive sessions may negatively affect mood states in professional athletes (Selmi et al., 2018). However, present investigated priming sessions were non-intensive and short-duration, and almost certainly do not negatively affect players' moods compared to highly intensive sessions. As such, they might be considered highly recommended and applicable in professional soccer settings.

The current results demonstrate potentially important performance benefits following morning priming sessions in men's professional soccer. However, the specific causative factor(s) could not be determined. Nevertheless, we can suggest that the positive effect of priming sessions on physical performance may be the result of acute hormonal changes. Typically, priming sessions aim to increase joint and muscle stiffness, activate motor neurons and elevate testosterone concentration (Harrison et al., 2019; Nutt et al., 2022). It is well known that testosterone concentration is positively associated with indices of explosive physical performance in trained populations (Cook & Crewther, 2012; Crewther et al., 2012). On the other hand, morning priming sessions have previously demonstrated efficacy in offsetting the circadian declines in testosterone concentration that are typically observed throughout the day (Cook et al., 2014; Nutt et al., 2022; Russell et al., 2016). Therefore, it is possible that the morning priming session had an impact on testosterone concentration, which consequently positively influenced players' high-intensity (i.e., explosive) efforts in the afternoon match. However, more research is needed to confirm this assertion as analysis of testosterone concentrations was not possible in the current study.

Previous research investigating technical performance and success in soccer have suggested that "successful" teams perform greater amounts of passes, shots, shots on target, and crosses compared to "unsuccessful" teams, highlighting technical performance as essential for competitive success (Lago-Peñas et al., 2011; Modric et al., 2022). A greater understanding of the strategies that can potentially be utilised to enhance players' technical performance during the match is therefore of great importance. The current study revealed that priming sessions implemented on morning match day did not enhance players' technical performance during the ensuing competitive match. While, a small improvement in the frequency of duels (small ES) in matches preceded by a priming session was evidenced, the number of passes, crosses, shots, tackles, and dribbles performed (all trivial to small ES) were similar irrespective of whether the priming session was implemented or not prior to play.

Although one can only speculate on causal factors, we are of the opinion that the specificities of the present priming sessions may be the reason for such findings. Specifically, it is well known that technical performance requires various cognitive, perceptual, and motor skills (Ali, 2011). While was

demonstrated that priming sessions generally may impact motor skills (Harrison et al., 2019; Nutt et al., 2022), their effect on perception and cognition skills which are important for the execution of technical performance is unknown and may explain the reason for the lack of enhancement of players' technical performance during matches. Irrespective of causality, it is noteworthy that application of priming session did not negatively affect technical performance, which is decisive in winning matches (Barnes et al., 2014; Moalla et al., 2018). This shows that the additional extra loading on match day will not have negative consequence on technical performances during competition strengthening the inclusion of a priming sessions prior to competition.

Several limitations should be noted when interpreting the findings of this study. The current data are reflective of the paradigms and practices of a single club, the training status of the players examined and the implemented priming sessions. Consequently, the results might only be generalisable to similar cohorts, levels of competition, and priming session prescriptions. Match observations preceded or not preceded by priming sessions were unequal in their sample sizes, and the discrepancy likely influenced the findings. Hence, future studies should be conducted to compare current findings using larger sample sizes. To ensure a more comprehensive understanding of the effect of priming sessions on physical performance, future studies should analyse additional factors associated with indices of physical performance such as muscle and deep-body temperature, testosterone concentrations, oxygen consumption, or heart rate (Nutt et al., 2022). This would help enable specific causative factors to be determined.

Conclusion

This study is the first to demonstrate that priming sessions may influence players' physical performance during competitive soccer matches. Specifically, greater total, moderate-, and high-intensity running distances were observed in matches preceded by a priming session than in matches not preceded by a priming session. Improvements in high-intensity accelerations and decelerations were also observed but with limited effect. This suggests that the application of a morning priming session may be an effective means of enhancing players' physical performance during matches. In contrast, no effect of the morning priming session was observed on technical performance. However, this result is not without interest as it shows that additional extra loading on match day did not negatively affect players' subsequent performances during the match.

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Data

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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